

AI-Based System for Detecting Fake Image Refund Claims in Instant Delivery Apps

a project report submitted by

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BONA FIDE CERTIFICATE

This is to certify that the project report entitled, “**AI-Based System for Detecting Fake Image Refund Claims in Instant Delivery Apps**” is a bonafide record of work of the following candidates who carried out the project work under my supervision during the academic year 2025-2026.

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ABSTRACT

With the rapid growth of digital media and online transactions, the authenticity of images has become a critical concern, particularly in domains such as refund verification and media authentication. The increasing ease of manipulating images using advanced editing tools and generative AI techniques has led to a surge in fraudulent activities, creating a strong need for reliable detection systems.

This project presents an AI-based web system designed to detect fake or manipulated images in real-time. The proposed solution integrates a hybrid deep learning architecture combining Convolutional Neural Networks (CNN) and Vision Transformers (ViT) to leverage both spatial feature extraction and global contextual understanding. The system is implemented using TensorFlow, ensuring scalability and efficient model deployment.

To support real-world applications, the system is optimized for low-latency processing, enabling fast and efficient analysis of images. It also incorporates concurrent user handling capabilities, making it suitable for web-based environments. The user interface is designed to be intuitive, allowing users to upload images and receive instant authenticity predictions.

The proposed system aims to provide a practical and reliable solution to verify image authenticity, reduce fraudulent claims, and enhance trust in digital media. Experimental results demonstrate improved classification accuracy and robust performance compared to traditional single-model approaches. Overall, this project contributes to AI-driven media forensics by delivering a scalable, efficient, and real-time fake image detection system.

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TABLE OF CONTENTS

Chapter No.	Title	Page No.
	ABSTRACT	iii
	ACKNOWLEDGEMENT	iv
	TABLE OF CONTENTS	v
	LIST OF FIGURES	vii
	LIST OF TABLES	viii
	LIST OF ABBREVIATIONS	ix
1	INTRODUCTION	1
	1.1 Background	1
	1.2 Objective	2
	1.3 Significance	2
2	LITERATURE REVIEW	4
3	METHODOLOGY	7
	3.1 Data Collection and Description	7
	3.2 Data Preprocessing	8
	3.2.1 Image Resizing	8
	3.2.2 Normalization	9
	3.2.3 Data Augmentation	9
	3.3 Model Architecture and Training	9
	3.3.1 CNN Branch – Detailed Explanation	10
	3.3.2 Vision Transformer (ViT)	11

	3.4 Overview of the Proposed System Architecture	13
	3.5 Feature Fusion Process	14
	3.6 Backend Processing and Inference Pipeline	16
	3.7 Frontend Interface and User Interaction	17
4	RESULTS AND DISCUSSION	18
	4.1 Performance Metrics	18
	4.2 Comparative Analysis	19
	4.2.1 Loss Curve Analysis	21
	4.2.2 Accuracy Curve Analysis	22
	4.3 Web Page Development	23
	4.3.1 Homepage Interface of FakeGuard AI	23
	4.3.2 Fake/AI-Generated Image Detection Interface	24
	4.3.3 Real Image Detection Result	25
	4.3.4 AI Assistant (Chatbot) Interface	26
	4.4 Ethical, Legal, and Privacy Considerations	27
	4.5 Data Leakage Prevention	28
	4.6 Impact of Training Data Size on Classification Accuracy	29
5	CONCLUSION	30
	5.1 Summary of Findings	30
	5.2 Limitations & Future Scope	31
	REFERENCES	32
	DECLARATION	34

LIST OF FIGURES

Figure No.	Title	Page No.
3.1	Model architecture and training	8
3.2	CNN layer of the architecture	11
3.3	VIT layer of the architecture	12
4.1	Confusion matrix of the Hybrid model	20
4.2	Training and Testing loss of the model	21
4.3	Training and Testing accuracy of the model	22
4.4	Homepage Interface of FakeGuard AI	23
4.5	Fake/AI-Generated Image Detection Interface	24
4.6	Real Image Detection Result	25
4.7	AI Assistant (Chatbot) Interface	26

LIST OF TABLES

Table No.	Title	Page No.
1	Component specifications and Parameters	13
2	Model specification of proposed model	15
3	Class-wise Performance metrics of fake and real image detection	19
4	Overall Performance metrics of the Hybrid model	19

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
ViT	Vision Transformer
DL	Deep Learning
ML	Machine Learning
ANN	Artificial Neural Network
RGB	Red Green Blue
API	Application Programming Interface
UI	User Interface
GPU	Graphics Processing Unit
CPU	Central Processing Unit
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
MLP	Multi-Layer Perceptron
GELU	Gaussian Error Linear Unit
ReLU	Rectified Linear Unit
GAP	Global Average Pooling
MHSA	Multi-Head Self-Attention
BN	Batch Normalization
LR	Learning Rate
AUC	Area Under Curve
ROC	Receiver Operating Characteristic

JSON

JavaScript Object Notation

REST

Representational State Transfer

CHAPTER 1

INTRODUCTION

1.1 Background

In today's digital era, images play a crucial role in communication, verification, and decision-making across various online platforms, especially in e-commerce and quick-commerce services. Many platforms rely on user-submitted images for refund claims, product complaints, and verification processes. However, with the rapid advancement of image editing software and AI-based image generation tools, it has become increasingly easy to manipulate or create highly realistic fake images. These manipulated images are often indistinguishable from genuine ones through manual inspection, leading to fraudulent refund claims and financial losses.

Traditional verification systems primarily depend on manual review or basic rule-based techniques, which are inefficient and incapable of detecting sophisticated image forgeries. As a result, there is a growing need for automated systems that can accurately identify fake or AI-generated images. Artificial Intelligence (AI) and Computer Vision techniques, particularly deep learning models, have emerged as powerful solutions for this problem.

Recent advancements in deep learning, especially Convolutional Neural Networks (CNN) and Vision Transformers (ViT), have significantly improved image classification and forgery detection capabilities. CNNs are effective in capturing local spatial features such as textures and edges, while Vision Transformers provide a global understanding of image context. Combining these models enables better detection of subtle inconsistencies in manipulated images. Therefore, AI-based fake image detection systems are becoming essential for enhancing trust, reducing fraud, and ensuring authenticity in digital platforms.

1.2 Objective

The main objective of this project is to design and develop a web-based AI system capable of detecting fake or manipulated images in real-time for refund verification and media authentication. The proposed system utilizes a hybrid deep learning model that combines Convolutional Neural Networks (CNN) and Vision Transformers (ViT) to improve detection accuracy by capturing both local and global image features. The system is implemented using TensorFlow and deployed as a web application to ensure accessibility and ease of use.

The system consists of several integrated components designed to enable efficient and accurate image authenticity detection. It begins with an intuitive web interface that allows users to upload images for analysis. Once an image is submitted, it undergoes preprocessing steps such as resizing, normalization, and conversion into tensor format to make it suitable for model input. Feature extraction is then performed using a Convolutional Neural Network (CNN), which captures local patterns and fine-grained details within the image. In parallel, a Vision Transformer (ViT) analyzes the global context to identify broader inconsistencies that may indicate manipulation. The features obtained from both CNN and ViT are fused to improve classification performance and robustness. Based on this combined analysis, the system provides a real-time prediction indicating whether the image is real or fake, along with a confidence score. The backend is implemented using a lightweight server to ensure fast and efficient inference, while also supporting concurrent user requests for scalability in real-world applications. Additionally, an optional chatbot interface can be integrated to assist users, provide guidance, and enhance overall usability.

The overall objective is to create an automated, accurate, and efficient system that enables users to instantly verify the authenticity of images in practical scenarios.

1.3 Significance

The proposed system provides a practical and intelligent solution for detecting fake images in real-world applications, particularly in refund verification systems. By automating the detection process, it eliminates the need for manual inspection and reduces the risk of human error.

The proposed system offers several key benefits in addressing image-based fraud detection. It helps prevent financial losses by identifying fake image-based refund claims, thereby strengthening fraud prevention mechanisms. The system provides real-time detection, enabling quick and efficient decision-making. Its high accuracy is achieved through the hybrid CNN–Vision Transformer (ViT) model, which effectively combines local and global feature learning.

Designed for scalability, the system can handle multiple users simultaneously in a web-based environment, making it suitable for large-scale applications. Additionally, it demonstrates practical deployment of artificial intelligence and deep learning techniques in real-world fraud detection scenarios. The user-friendly interface ensures ease of use, allowing even non-technical users to upload images and obtain results effortlessly.

From a business perspective, it adds value to e-commerce platforms by automating verification processes and enhancing customer trust. Furthermore, the system contributes to ongoing research in AI-based media forensics and image authentication. Its low-latency performance ensures fast processing, making it well-suited for real-time applications.

Overall, this project contributes to the field of AI-driven image forensics by providing a scalable, efficient, and user-friendly solution for fake image detection. It bridges the gap between advanced deep learning models and real-world applications, enhancing trust and reliability in digital transactions.

CHAPTER 2

LITERATURE REVIEW

Fake image detection has emerged as a significant research area within Computer Vision and Artificial Intelligence due to the rapid advancement of image manipulation and AI-generated content. Early approaches relied on traditional image forensics techniques such as error level analysis (ELA), noise pattern analysis, and handcrafted feature extraction methods [1]. Classical machine learning algorithms like Support Vector Machines (SVM), Decision Trees, and k-Nearest Neighbors (k-NN) were used along with features such as color inconsistencies, compression artifacts, and pixel-level irregularities [2]. While these methods showed reasonable performance for basic image tampering detection, they struggled to generalize across different manipulation techniques and failed to detect highly realistic AI-generated images [2].

The introduction of deep learning brought a major shift in fake image detection. Convolutional Neural Networks (CNNs) enabled automatic feature extraction from images, significantly improving detection accuracy [3]. CNN-based models could capture spatial hierarchies such as edges, textures, and fine-grained artifacts [3]. However, CNNs primarily focus on local features and often fail to capture global contextual inconsistencies present in manipulated images.

To overcome these limitations, Vision Transformers (ViT) were introduced, leveraging self-attention mechanisms to understand global relationships within an image. Unlike CNNs, transformers process the entire image context simultaneously, enabling better detection of subtle inconsistencies and unnatural patterns [4]. Combining CNN and Vision Transformer architectures allows models to utilize both local feature extraction and global contextual understanding, resulting in improved performance for fake image classification [4].

Contribution of Existing Literature

1. Superiority of Deep Learning over Traditional Methods

Several studies have demonstrated that deep learning models significantly outperform traditional machine learning approaches in fake image detection.

CNN-based models can effectively learn discriminative features to distinguish real and fake images, achieving higher accuracy compared to classical techniques.

However, many of these approaches primarily focus on improving model performance and do not address real-world deployment challenges [1].

2. Hybrid Deep Learning Approaches

Hybrid models combining multiple techniques have shown improved robustness in detecting manipulated images. Systems that integrate deep learning with traditional forensic features demonstrate enhanced detection accuracy across various manipulation methods. In particular, combining CNN and Vision Transformer models improves performance by capturing both local spatial details and global contextual relationships, making detection more reliable [12].

3. Transformer-Based Image Analysis

Recent research highlights the effectiveness of transformer-based models in image forensics. Vision Transformers utilize self-attention mechanisms to analyze global dependencies within an image, enabling better detection of inconsistencies that may not be visible through local feature analysis alone. These models also contribute to improved interpretability by identifying regions of interest within manipulated images [3].

4. Deepfake and AI-Generated Image Detection

With the rise of generative models, detecting AI-generated images has become increasingly challenging. Deep learning-based forensic techniques have been explored to identify deepfake content and synthetic images. These studies emphasize the importance of robust detection systems for ensuring privacy, authenticity, and trust in digital media [4].

5. Application-Oriented Deep Learning Systems

While many studies focus on improving detection accuracy, fewer works address real-world implementation.

There is a growing need to integrate deep learning models into practical systems such as web-based applications [5].

6. Generalization and Real-World Challenges

Existing literature also identifies challenges such as dataset limitations, generalization across different manipulation techniques, and computational complexity. Although advanced models achieve high accuracy, real-world deployment requires balancing performance with efficiency. Systems must also handle multiple users, ensure low-latency responses, and maintain reliability under practical conditions [6].

Summary

Overall, the literature shows that traditional methods provided the foundation for fake image detection, but deep learning and transformer-based approaches have significantly improved accuracy and robustness. Hybrid architectures combining CNN and Vision Transformers represent a powerful solution by leveraging both local and global feature learning. However, a gap still exists in deploying these models in real-time, scalable systems. This project addresses that gap by developing an efficient, web-based fake image detection system suitable for practical applications such as refund verification and media authentication.

CHAPTER 3

METHODOLOGY

3.1 Data Collection and Description

The dataset used in this project is the CIFAKE dataset, which contains a large collection of both real-world images and artificially generated (fake) images. The dataset was accessed using automated tools within Google Colab, ensuring efficient downloading and management.

The images are categorized into:

- Real Images → captured using cameras
- Fake Images → generated using AI models or manipulated

The dataset is further divided into:

- Training dataset (used to train models)
- Testing dataset (used to evaluate performance)

This dataset is highly relevant for real-world applications such as:

- Refund fraud detection
- Media authentication
- Deepfake detection

The use of both real and synthetic images helps the model learn diverse patterns, improving generalization and robustness.

All images are RGB format with a resolution of 32×32 pixels, which reduces computational complexity and allows faster model training. The balanced nature of the dataset ensures that the model does not develop bias toward any specific class.

This dataset is highly relevant for real-world applications such as:

- Fraud detection in refund systems
- Media authenticity verification
- Deepfake detection

The diversity and balance of the dataset improve the model's ability to generalize to unseen data.

3.2 Data Preprocessing

Data preprocessing is a crucial step in preparing raw image data for deep learning models. Since images may vary in quality, lighting, and orientation, preprocessing ensures consistency and improves model performance.

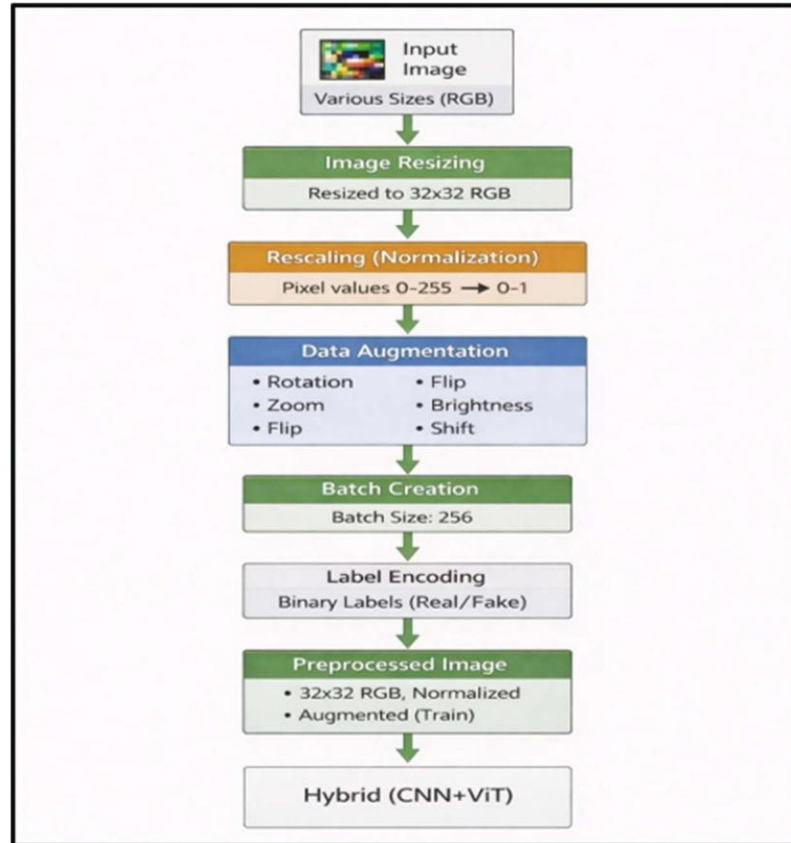


Fig 3.1 : Model architecture and training

3.2.1 Image Resizing

Image resizing is performed to ensure that all input images have a uniform spatial dimension compatible with the model architecture. In this work, all images are resized to 32×32 pixels with 3 color channels (RGB). This standardization is essential because deep learning models, particularly convolutional neural networks and vision transformers, require fixed-size inputs for efficient batch processing. Additionally, resizing reduces computational complexity and memory usage while maintaining sufficient visual information for feature extraction. This ensures compatibility with the model architecture and reduces computational overhead.

3.2.2 Normalization

Normalization is applied to scale the pixel intensity values of input images to a consistent range, which is essential for stable and efficient model training. In this work, normalization is performed by rescaling the pixel values using a factor of $1/255$, thereby converting the original range of $[0, 255]$ to $[0, 1]$. This transformation reduces the magnitude of input values, helping to prevent large gradient updates and improving numerical stability during optimization. As a result, the model converges faster and learns more effectively, particularly when using optimizers such as AdamW. Normalization also ensures that all input features contribute proportionately to the learning process, leading to improved generalization performance on unseen data.

3.2.3 Data Augmentation

Data augmentation is applied to the training dataset to increase data diversity and improve the model's ability to generalize to unseen samples. In this work, augmentation is performed using several random transformations, including rotation (up to 15 degrees), zooming (up to 15%), horizontal flipping, brightness adjustment (range 0.85 to 1.15), and width and height shifting (up to 10%). These transformations simulate real-world variations such as changes in orientation, lighting conditions, and object positioning, allowing the model to learn more robust and invariant features. By exposing the model to multiple altered versions of the same image, data augmentation helps reduce overfitting and enhances overall performance. It is applied only during training, while validation and test datasets remain unaltered except for normalization, ensuring a fair evaluation of the model.

3.3 Model Architecture and Training

The core of the proposed system is a Hybrid Deep Learning Architecture that combines the strengths of Convolutional Neural Networks (CNN) and Vision Transformers (ViT). This combination is used to overcome the limitations of individual models and achieve better performance in fake image detection.

CNNs are highly effective in extracting local spatial features such as edges, textures, and small artifacts, while Vision Transformers are designed to capture global contextual relationships within an image. By integrating both approaches, the model gains a comprehensive understanding of the image at both local and global levels.

The architecture consists of three major components:

- CNN Branch → Local Feature Extraction
- Vision Transformer Branch → Global Context Understanding
- Feature Fusion Layer → Final Classification

The input image is processed in parallel by both branches, and their outputs are combined for final prediction.

3.3.1 CNN Branch – Detailed Explanation

The Convolutional Neural Network (CNN) in the proposed hybrid model is responsible for extracting local spatial features from input images. It processes the input image of size $32 \times 32 \times 3$ (RGB) through a sequence of convolutional and normalization layers to learn hierarchical representations such as edges, textures, and patterns.

Initially, the input image is passed through the first convolutional layer, which consists of 32 filters of size 3×3 with ReLU activation. This layer scans the image using learnable kernels, producing feature maps that capture low-level features like edges and color gradients. The use of padding (“same”) ensures that the spatial dimensions of the image remain unchanged after convolution. Following this, Batch Normalization is applied to stabilize the learning process by normalizing the activations, reducing internal covariate shift and improving convergence speed.

The output is then fed into a second convolutional layer with 64 filters of size 3×3 , again followed by Batch Normalization. This deeper layer captures more complex and abstract features by combining the low-level features learned in the previous layer. As the number of filters increases, the network gains the ability to represent richer feature patterns present in the image.

After the convolutional operations, a Global Average Pooling (GAP) layer is applied. Instead of flattening the entire feature map, GAP computes the average value of each feature map, resulting in a 64-dimensional feature vector. This significantly reduces the number of parameters, prevents overfitting, and ensures that the model focuses on the presence of features rather than their exact spatial locations. Overall, the CNN component acts as a feature extractor, capturing local and spatial information efficiently. These extracted features are later combined with the global contextual features obtained from the Vision Transformer (ViT), enabling the hybrid model to leverage both local and global representations for improved classification performance.

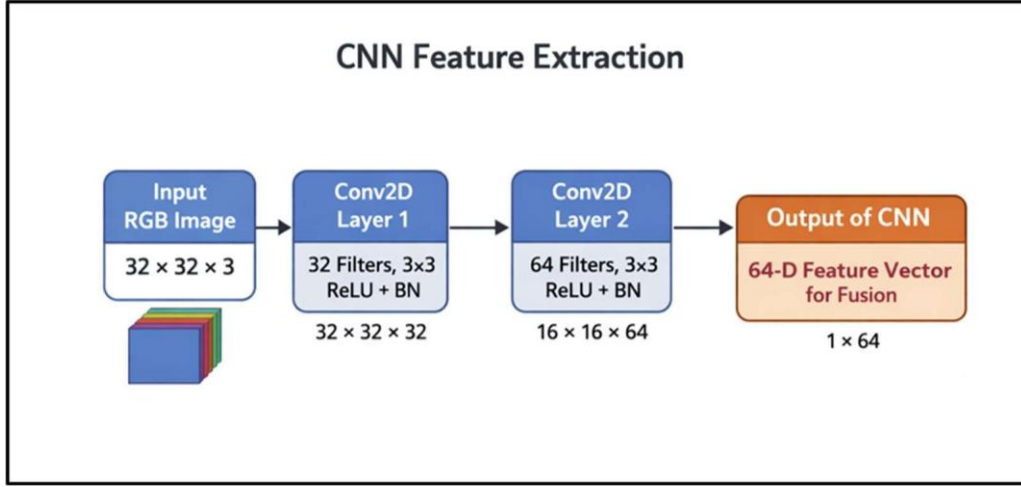


Fig 3.2 : *CNN layer of the architecture*

3.3.2 Vision Transformer (ViT)

The Vision Transformer (ViT) component in the proposed hybrid model is designed to capture global contextual relationships within the input image, complementing the local feature extraction performed by the CNN. Unlike convolutional networks, which focus on localized receptive fields, the ViT processes the image as a sequence of patches and models long-range dependencies using self-attention mechanisms.

Initially, the input image of size $32 \times 32 \times 3$ is divided into smaller non-overlapping patches of size 4×4 . This results in a total of 64 patches, where each patch is flattened into a vector of size 48 ($4 \times 4 \times 3$). These patch vectors are then passed through a linear projection layer, which maps each patch into a higher-dimensional embedding space of 64 dimensions.

To retain spatial information, positional embeddings are added to the projected patch embeddings. This step is crucial because transformers do not inherently encode positional relationships. The resulting sequence of embedded patches is then fed into a stack of Transformer encoder layers.

Each Transformer encoder layer consists of two main subcomponents: Multi-Head Self-Attention (MHSA) and a Feed-Forward Neural Network (MLP block). In the self-attention mechanism, each patch attends to all other patches in the image, allowing the model to learn global dependencies and contextual relationships. The use of multiple attention heads (4 heads in this model) enables the network to capture different types of relationships

simultaneously. The output of the attention layer is combined with the input using residual connections, followed by Layer Normalization to stabilize training. The second subcomponent, the MLP block, consists of fully connected layers with GELU activation and dropout for regularization. This block further refines the learned representations. These two subcomponents are repeated for 4 transformer layers, enabling deeper feature learning.

After passing through all transformer layers, the final sequence of encoded patches is normalized and flattened, resulting in a high-dimensional feature vector. This vector is then passed through a dense layer to reduce its dimensionality and prepare it for fusion with CNN features. Overall, the Vision Transformer effectively captures global structure and long-range dependencies in the image, which enhances the model's ability to understand complex patterns. When combined with the CNN's local feature extraction, the hybrid architecture achieves improved performance in image classification tasks.

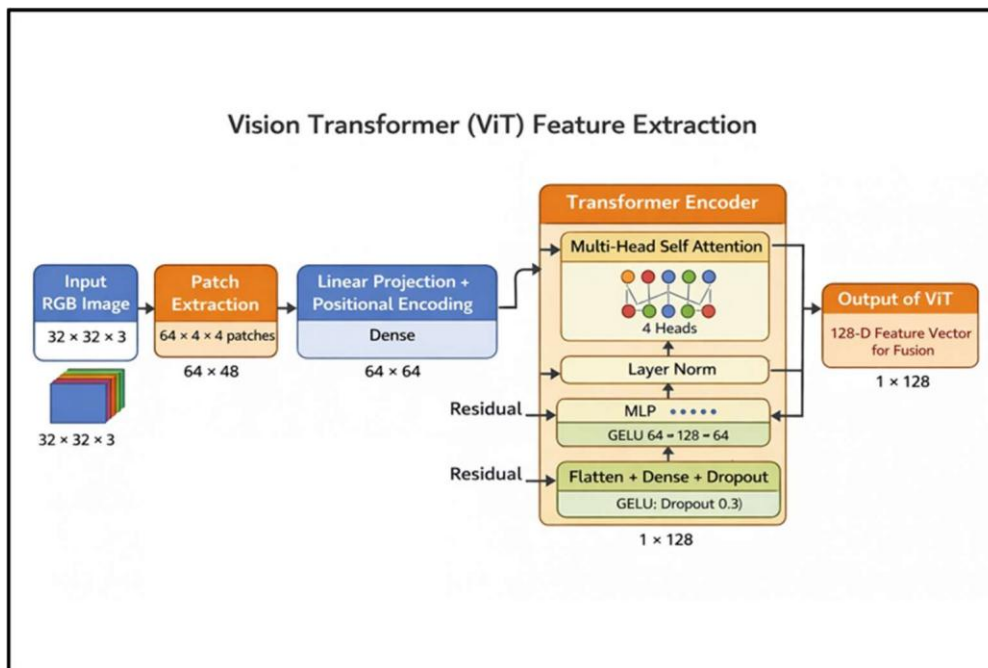


Fig 3.3 : *ViT layer of the architecture*

3.4 Overview of the Proposed System Architecture

The proposed system is designed based on a client–server architecture, integrating a web-based frontend, a backend server, and a deep learning inference unit.

The frontend is responsible for user interaction, allowing users to upload images and view the classification results. It is developed using standard web technologies, ensuring accessibility across different devices and platforms.

The backend server is implemented using a lightweight web framework, which acts as an intermediary between the user interface and the deep learning model. It handles tasks such as receiving user requests, performing image preprocessing, and managing communication with the trained model.

The deep learning inference unit processes the pre-processed images using the hybrid CNN–Vision Transformer model to generate predictions. By separating the frontend, backend, and model components, the system achieves better modularity.

This separation of responsibilities allows independent handling of:

- User requests
- Image preprocessing
- Model inference

As a result, the system demonstrates improved maintainability, scalability, and performance.

Table 1 : Component specifications and Parameters

Component	Specification & Parameters
Input Layer	Image size: $32 \times 32 \times 3$ (RGB input)
Data Preprocessing	Rescaling ($1/255$), augmentation: rotation (15°), zoom (0.15), horizontal flip, brightness (0.85–1.15), width/height shift (0.1)
CNN Block 1	Conv2D (32 filters, 3×3 , ReLU) + Batch Normalization
CNN Block 2	Conv2D (64 filters, 3×3 , ReLU) + Batch Normalization
Feature Extraction (CNN)	Global Average Pooling $\rightarrow$ 64-dimensional feature vector
Patch Extraction	Patch size: $4 \times 4 \rightarrow$ Total patches: 64
Patch Encoding	Dense projection (64-dim) + positional embedding

Transformer Encoder	4 layers, 4 heads, key_dim = 16, LayerNorm + MLP (128 $\rightarrow$ 64), Dropout (0.1)
Activation Functions	ReLU (CNN layers), GELU (Transformer & MLP), Sigmoid (output layer)
Optimization & Training	Optimizer: AdamW (weight decay = 1e-4), Learning rate: 3e-4 with CosineDecayRestarts, Loss: Binary Crossentropy, Metrics: Accuracy & AUC, Epochs: 25, Batch size: 256

3.5 Feature Fusion Process

The feature fusion process plays a crucial role in the proposed hybrid architecture by integrating the complementary representations learned from the CNN and Vision Transformer (ViT) branches. The CNN branch is specialized in capturing low-level and mid-level local features, such as edges, textures, and spatial patterns, which are essential for identifying fine-grained details in images. After passing through convolutional and normalization layers, these features are compressed into a 64-dimensional vector using Global Average Pooling, ensuring a compact and noise-reduced representation.

In parallel, the ViT branch processes the same input image by dividing it into patches and modeling global dependencies using self-attention mechanisms. This enables the model to understand relationships between distant regions of the image, capturing semantic and structural information that CNNs may miss. After multiple transformer encoding layers and a dense projection, the ViT produces a 128-dimensional feature vector that encodes high-level contextual information.

To combine these heterogeneous feature representations, a feature-level fusion strategy using concatenation is employed. The CNN and ViT feature vectors are concatenated along the feature dimension to form a unified 192-dimensional feature vector. This operation preserves the full information from both branches without any loss, allowing the model to utilize both local precision and global context simultaneously.

Following fusion, the combined feature vector is passed through a Multi-Layer Perceptron (MLP) head, which consists of fully connected layers with GELU activation functions and dropout regularization (0.3). This stage performs non-linear transformation and feature refinement, enabling the model to learn complex interactions between local and global features. The final layer uses a sigmoid activation function to output a probability score for binary classification. This fusion approach significantly enhances the model's representational capacity by bridging the gap between spatial feature extraction (CNN) and contextual understanding (ViT). As a result, the hybrid model achieves improved robustness, better generalization, and higher classification accuracy compared to standalone architectures.

Table 2 : Model specification of proposed model

Model Parameter	Trainable Parameters
Conv2D Layer 1	896
Conv2D Layer 2	18,496
Patch Encoder	7,232
Multi-Head Attention (per layer)	$16,640 \times 4 \approx 66,560$
Transformer MLP Blocks	$\sim 65,000$
Dense Layer (ViT Flatten)	524,416
CNN Feature Layer (GAP)	0
Fusion Dense Layer	24,704
Classification Head (MLP)	$\sim 8,256$

3.6 Backend Processing and Inference Pipeline

The The backend processing and inference pipeline is responsible for executing the core functionality of the proposed CNN+ViT hybrid model and managing the complete flow from input acquisition to result generation. It acts as the central processing unit of the system, handling incoming requests from the frontend, performing necessary preprocessing, executing the trained model, and returning the final predictions in a structured format. The backend is implemented using a server-side framework that exposes RESTful APIs to facilitate communication between the user interface and the model.

When a user uploads an image or video, the backend first validates the input by checking file format, size constraints, and data integrity. For images, the input is converted into a suitable format and undergoes preprocessing steps such as resizing, normalization, and tensor conversion. For videos, the backend extracts frames at regular intervals and processes each frame individually. These preprocessed inputs are then passed to the trained CNN+ViT hybrid model, where the CNN component extracts local spatial features and the Vision Transformer captures global contextual relationships.

These outputs are then post-processed to determine the final prediction label (Real or Fake) along with associated confidence scores. In the case of video inputs, predictions from multiple frames are aggregated using a voting or averaging mechanism to produce a single, robust decision. The backend then formats these results into a structured JSON response, which includes prediction labels, confidence values, and additional metadata. Furthermore, the backend is designed to handle multiple concurrent requests efficiently, ensuring scalability and responsiveness of the system. Techniques such as optimized model loading, GPU utilization (if available), and asynchronous request handling are employed to improve performance. Error handling mechanisms are also incorporated to manage invalid inputs or processing failures gracefully..

3.7 Frontend Interface and User Interaction

The frontend interface is designed to provide an intuitive, responsive, and user-friendly platform for interacting with the proposed CNN+ViT hybrid detection system. It acts as the bridge between the user and the backend inference engine, enabling seamless submission and analysis of images and videos. The interface typically includes dedicated sections such as a home page, detection module, and chatbot support, allowing users to easily navigate through different functionalities. Users can upload image or video files through a simple drag-and-drop mechanism or file selection option, with clear guidelines on supported formats and file size limits.

Once a file is uploaded, the frontend communicates with the backend through RESTful API calls, sending the input data for preprocessing and prediction using the hybrid CNN+ViT model. During this process, the interface provides real-time feedback through visual indicators such as loading animations or progress messages, ensuring that users are informed about the system status. After processing, the results are displayed in a clear and structured format, including the predicted class label (Real or Fake), confidence scores, and additional details such as frame-based analysis for videos.

The chatbot assists users by answering queries related to how the system works, how to upload files, and how to interpret results. This feature enhances accessibility, especially for non-technical users, by providing instant guidance and improving overall usability. The frontend is also designed to handle multiple user interactions efficiently, ensuring smooth performance even when multiple requests are made. It emphasizes responsiveness and compatibility across different devices and screen sizes, thereby offering a consistent user experience. ensures that the results are presented in an understandable and visually appealing manner, making the system practical and user-centric.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Performance Metrics

The performance of the proposed hybrid CNN–Vision Transformer (ViT) model is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of the model’s effectiveness in distinguishing between real and fake images.

Accuracy represents the overall correctness of the model’s predictions, and the high accuracy achieved indicates stable and reliable classification performance. Precision measures the proportion of correctly identified positive predictions, while recall evaluates the model’s ability to detect all relevant positive instances.

The consistently high precision and recall values demonstrate that the model effectively minimizes both false positives and false negatives. Furthermore, the F1-score, which is the harmonic mean of precision and recall, reflects a balanced trade-off between these two metrics. Overall, the performance metrics confirm that the proposed hybrid model achieves strong classification capability and maintains reliable detection performance across different test scenarios.

Accuracy

Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted samples out of the total samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Precision indicates how many of the predicted positive samples are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

F1-Score

F1-score is the harmonic mean of precision and recall, providing a balance between them.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Recall

Recall measures how many actual positive samples are correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

Table 3 : *Class-wise Performance metrics of fake and real image detection*

Class	Precision	F1-Score	Recall	Support
Fake	0.91	0.96	0.93	10000
Real	0.96	0.90	0.93	10000

Table 4 : *Overall Performance metrics of the Hybrid model*

Metrics	Value	Support
Accuracy	0.93	20000
Macro Average	Precision – 0.93	Recall – 0.93
Weighted Average	Precision 0.93	Recall – 0.93

4.2 Comparative Analysis

Based on the evaluation metrics and result visualizations, the proposed hybrid CNN–Vision Transformer (ViT) model demonstrates strong overall performance and effectiveness in fake image detection.

The confusion matrix further validates the model’s performance by showing minimal misclassification between classes.

These results indicate that the model achieves high true positive and true negative rates. The relatively low number of false negatives highlights the model’s strong ability to detect fake images, which is critical in fraud detection scenarios. Although there are some false positives, the model prioritizes identifying fake images accurately, reducing the risk of fraudulent cases being accepted.

Furthermore, the training and validation loss curves show a smooth and consistent decrease over the epochs, indicating stable learning and effective convergence. At the same time, the training and validation accuracy curves exhibit a steady upward trend and remain closely aligned, suggesting that the model generalizes well to unseen data without significant overfitting.

Overall, the comparative analysis confirms that the proposed hybrid architecture provides reliable, robust, and balanced performance, making it suitable for real-time fake image detection applications.

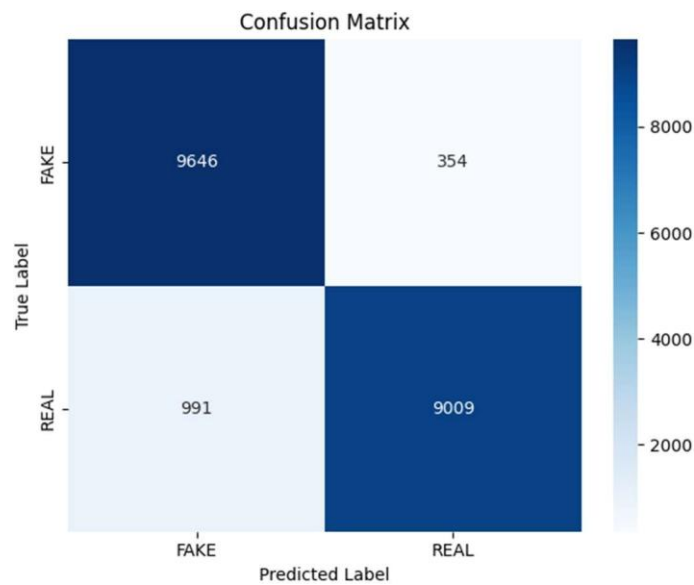


Fig 4.1 : *Confusion matrix of the Hybrid model*

4.2.1 Loss Curve Analysis

The loss curve provides insight into the learning behavior and convergence of the model during training. The training loss shows a steady decreasing trend, indicating that the model is effectively learning patterns from the training data.

The validation loss, although showing minor fluctuations across epochs, follows an overall downward trend. This behavior suggests that the model is able to generalize well to unseen data while maintaining stability during training.

The temporary variations observed in the validation loss can be attributed to differences in validation samples; however, there is no significant divergence between training and validation loss. This indicates that the model does not suffer from overfitting.

Overall, the loss curve confirms that the model achieves stable convergence and maintains a good balance between learning and generalization.

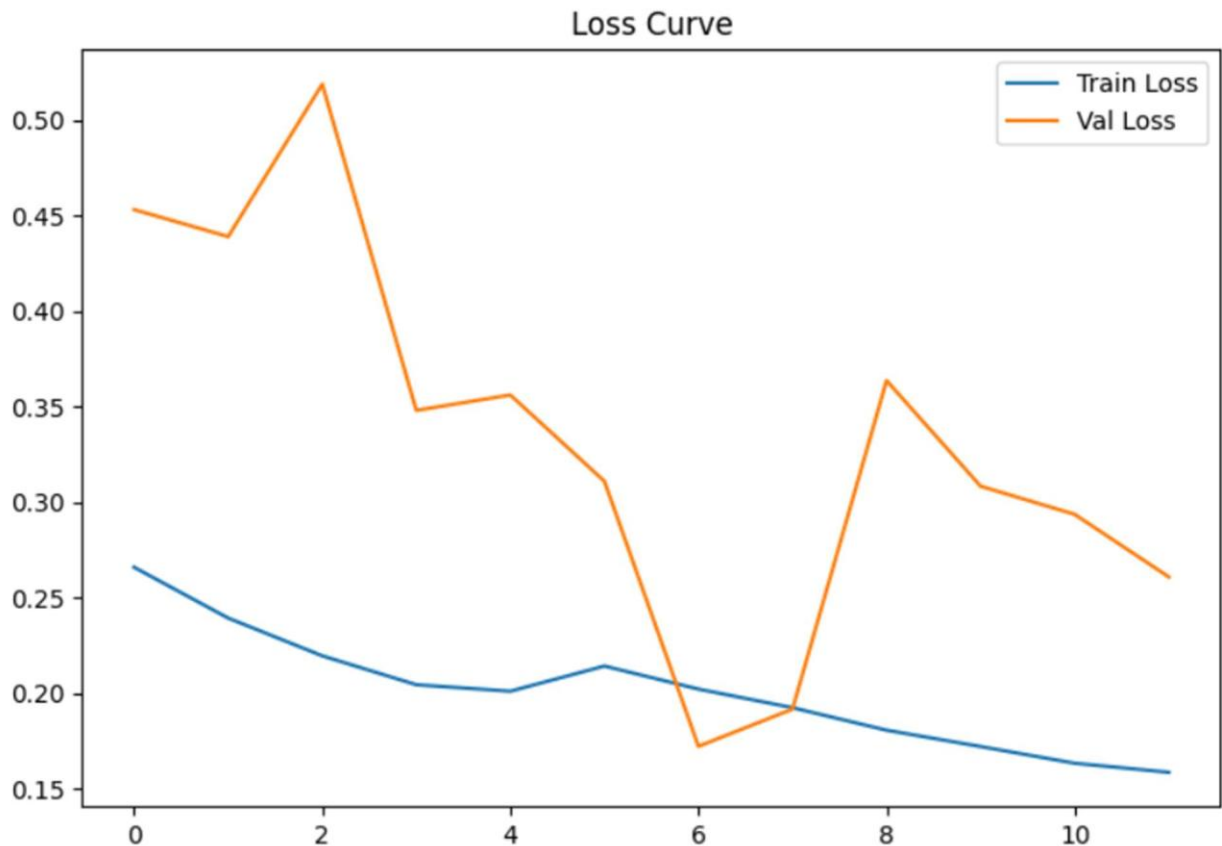


Fig 4.2 : *Training and Testing loss of the model*

4.2.2 Accuracy Curve Analysis

The accuracy curve illustrates the model's performance improvement over the training epochs. The training accuracy shows a steady and consistent increase, indicating that the model is effectively learning patterns from the training data.

The validation accuracy also demonstrates an overall upward trend, although with minor fluctuations at certain epochs. These fluctuations are common in deep learning models and are typically caused by variations in validation data. Despite these variations, the validation accuracy remains close to the training accuracy, indicating strong generalization capability.

At later epochs, both training and validation accuracy converge, with the model achieving approximately 93–94% accuracy. This convergence confirms that the model has reached a stable learning state.

The close alignment between training and validation accuracy curves suggests that the model does not suffer from significant overfitting or underfitting. Instead, it maintains a good balance between learning and generalization. The combined analysis of confusion matrix, loss curve, and accuracy curve confirms that the proposed hybrid CNN–ViT model achieves stable convergence, high accuracy, and robust generalization performance.

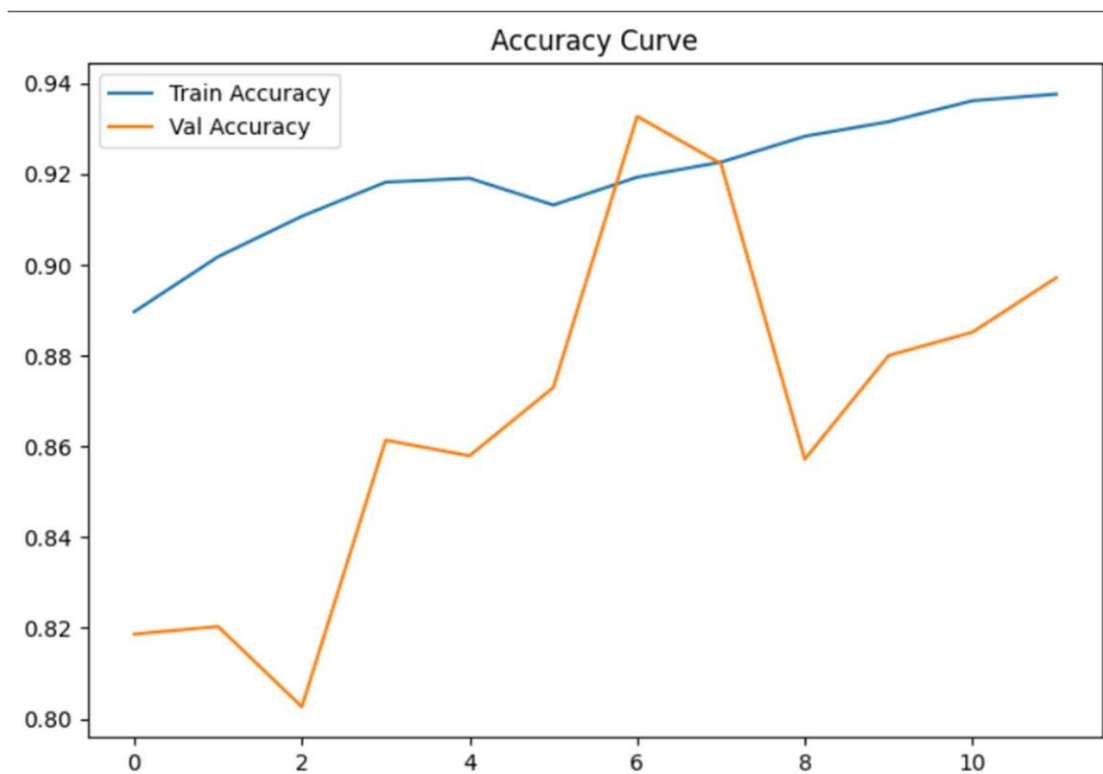


Fig 4.3 : *Training and Testing accuracy of the model*

4.3 Web Page Development

4.3.1 Homepage Interface of FakeGuard AI

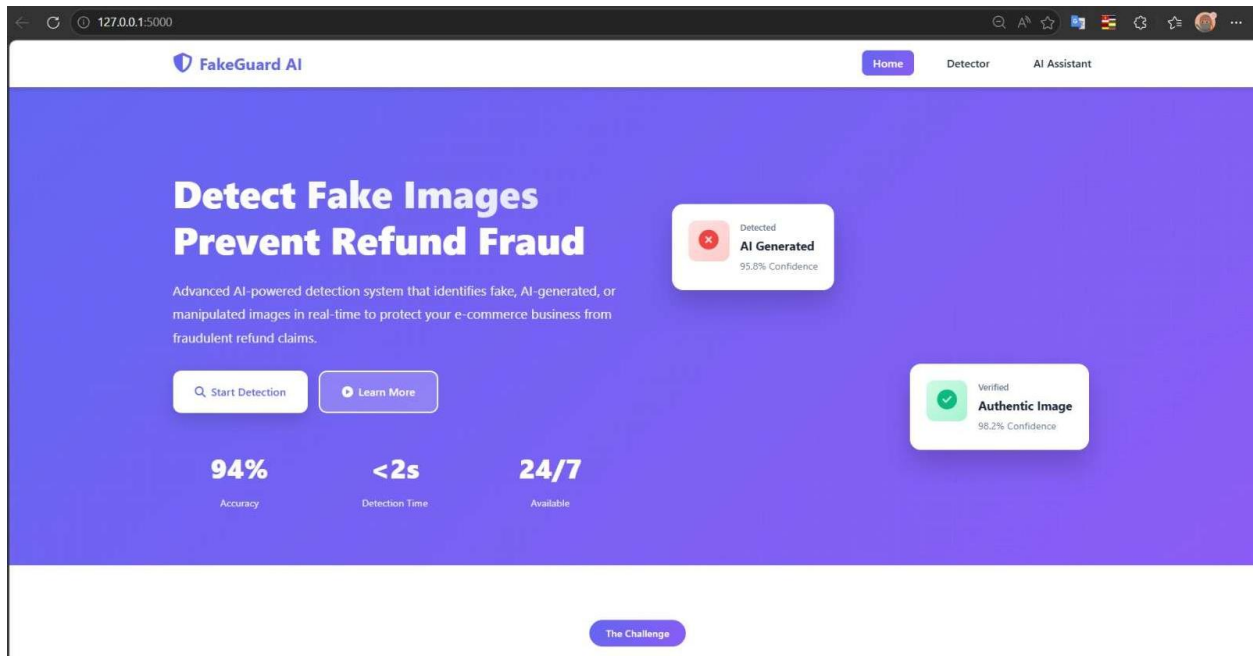


Fig 4.4 : *Homepage Interface of FakeGuard AI*

This figure represents the homepage of the FakeGuard AI system, which serves as the entry point of the application. The interface is designed using modern web technologies with a visually appealing gradient layout.

The page clearly communicates the purpose of the system, which is to detect fake images and prevent refund fraud. It also highlights important system metrics such as accuracy, detection speed, and availability.

Interactive navigation elements such as Home, Detector, and AI Assistant allow users to seamlessly move between modules. Action buttons like *Start Detection* guide users toward the main functionality of the system.

The homepage emphasizes clarity, usability, and accessibility, ensuring that users can quickly understand the purpose and functionality of the application.

4.3.2 Fake/AI-Generated Image Detection Interface

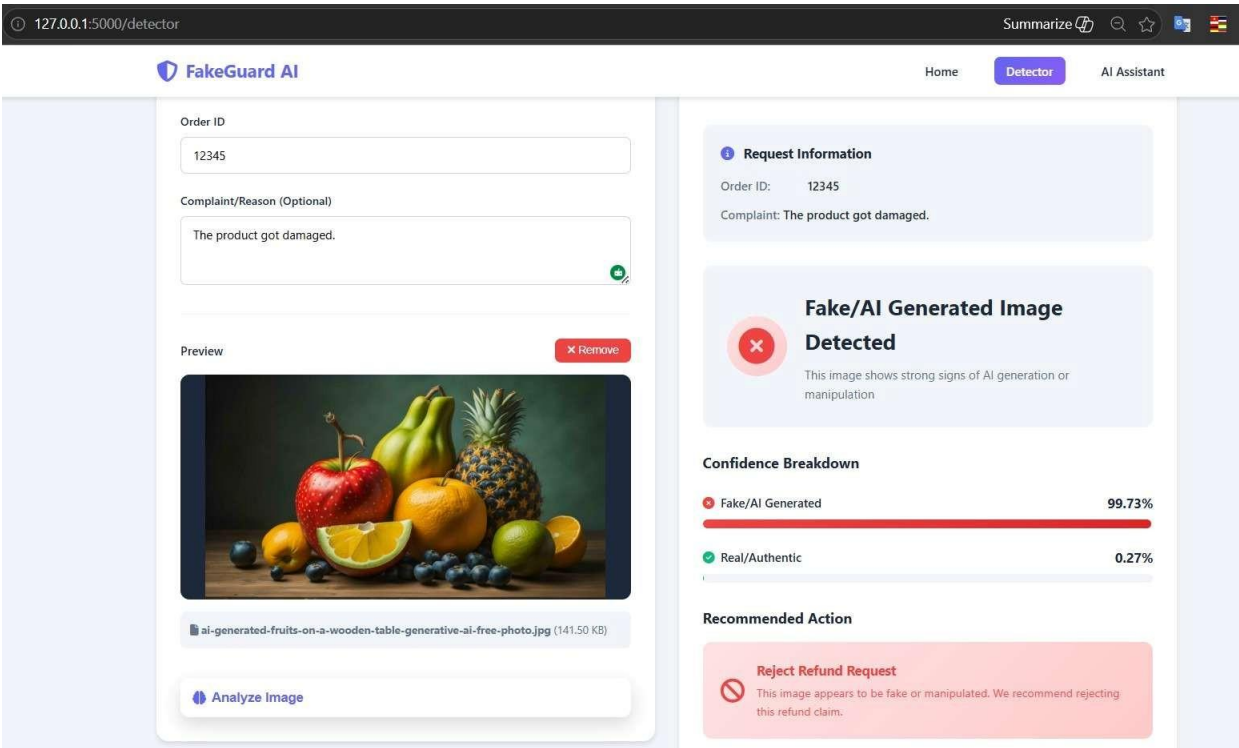


Fig 4.5 : *Fake/AI-Generated Image Detection Interface*

This figure shows the output when an AI-generated or manipulated image is uploaded. The system correctly classifies the image as “Fake/AI Generated Image Detected.” The confidence breakdown clearly indicates a high probability for the fake class, demonstrating the effectiveness of the hybrid model in identifying synthetic content. A red warning indicator is used to highlight the detection result, making it easy for users to interpret. The interface also provides a recommended action, suggesting rejection of the refund request. The layout maintains consistency with the authentic case, ensuring a uniform user experience. The use of color-coded indicators enhances clarity and supports quick decision-making. Additionally, the system provides a structured confidence visualization, allowing users to understand the degree of certainty in the prediction. This transparency improves trust in the system’s output. The fast response time further ensures that decisions can be made instantly, which is critical in real-time fraud detection scenarios. Overall, this result demonstrates the system’s strong capability to accurately identify manipulated images and assist users in preventing fraudulent refund claims.

4.3.3 Real Image Detection Result

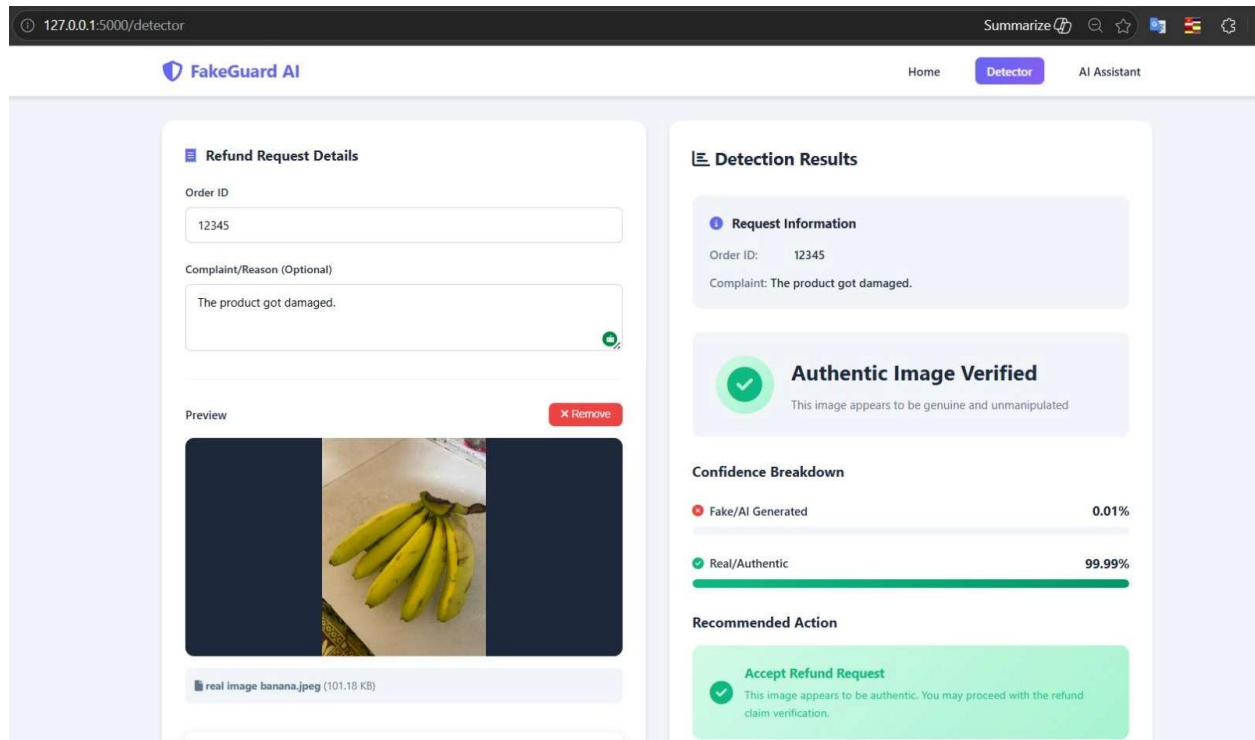


Fig 4.6 : *Real Image Detection Result*

This figure illustrates the system output when a real image is processed by the FakeGuard AI system.

After the user uploads the image through the detector module, the system performs preprocessing and passes the image to the hybrid CNN–Vision Transformer model for classification. The model successfully identifies the image as “Real/Authentic Image.”

The interface displays:

- A green verification indicator, representing authenticity
- A detailed confidence breakdown, showing a very high probability for the real class
- The uploaded image preview along with user-provided details such as order ID and complaint

In addition, the system generates a recommended action, suggesting that the refund request can be accepted. This highlights the practical application of the system in real-world refund verification scenarios.

The consistent layout and clear visualization ensure that users can easily interpret the results without requiring technical expertise.

4.3.4 AI Assistant (Chatbot) Interface

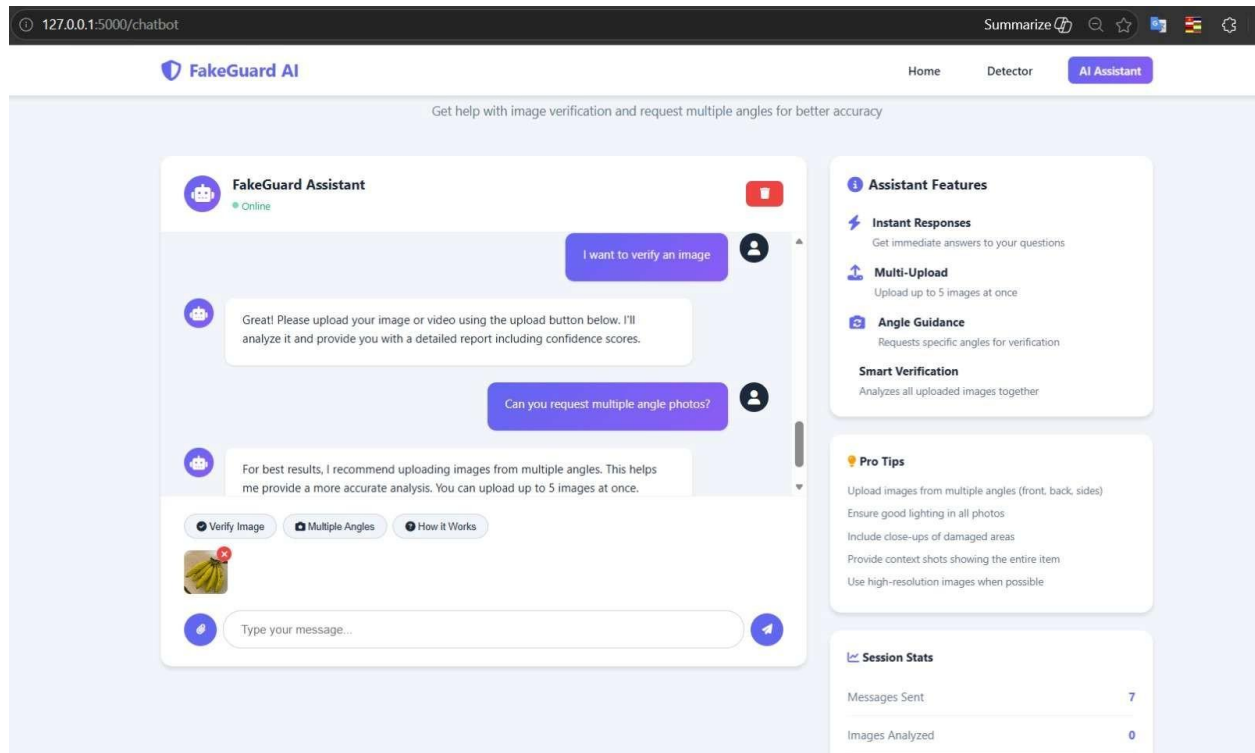


Fig 4.7 : AI Assistant (Chatbot) Interface

This figure illustrates the AI Assistant module integrated into the FakeGuard AI system, designed to enhance user interaction and provide intelligent guidance during the image verification process.

The chatbot interface enables real-time communication between the user and the system. Users can interact with the assistant by asking queries or requesting help related to image verification. The chatbot responds instantly with relevant suggestions, improving the usability of the system. The interface includes features such as:

- Instant responses to user queries
- Multi-image upload support, allowing users to upload multiple images for better verification
- Angle guidance, recommending users to upload images from different perspectives
- Smart verification, where multiple inputs are analyzed collectively for improved accuracy

Additionally, a “Pro Tips” section is displayed on the right side, providing useful recommendations such as ensuring proper lighting, capturing multiple angles, and including close-up shots.

4.4 Ethical, Legal, and Privacy Considerations

The proposed system is developed with careful consideration of ethical, legal, and privacy aspects associated with AI-based image verification. All images used in the experimental study are obtained from open-source datasets, ensuring compliance with data usage guidelines and avoiding unauthorized data access.

Since the system processes user-uploaded images to determine authenticity, it is essential to ensure secure handling of data. Images are stored only temporarily during processing and can be removed immediately after inference to minimize privacy risks. This approach reduces the possibility of data misuse and enhances user trust.

Fairness and transparency are critical factors in this system, as it involves automated decision-making that can influence refund outcomes. The possibility of misclassification is acknowledged, and therefore, the system is designed to provide confidence scores and clear outputs to improve interpretability and accountability.

In addition, the system aims to avoid bias by training on a balanced dataset containing both real and AI-generated images. This helps ensure that predictions are not skewed toward a particular class. Regular monitoring and evaluation of model performance are also necessary to maintain fairness over time.

Although the current implementation operates on a local host environment, real-world deployment would require additional safeguards. These include:

- Secure communication protocols (e.g., HTTPS)
- Restricted data access mechanisms
- Authentication and authorization controls
- Compliance with relevant data protection regulations

Furthermore, logging and auditing mechanisms can be incorporated to track system usage and detect potential misuse. Implementing encryption techniques for data storage and transmission can further enhance security.

The responsible use of both real and AI-generated datasets is emphasized to ensure ethical integration into digital commerce platforms. Transparency in AI-driven decision-making processes plays a key role in maintaining user confidence and promoting responsible adoption of the system.

4.5 Data Leakage Prevention

To ensure accurate performance evaluation and prevent data leakage, strict dataset preparation and splitting strategies were followed during the development of the model.

The dataset was divided into training and testing sets prior to any preprocessing steps, thereby avoiding the introduction of statistical bias. This ensures that the model does not gain unintended information from the test data during training.

To further prevent leakage, images were split at the source level, ensuring that visually similar or augmented versions of the same image did not appear in both training and testing sets. This step is crucial in avoiding the model learning sample-specific patterns rather than generalizable features.

Each subset was processed independently, including:

- Image resizing
- Normalization
- Tensor conversion

Normalization parameters were calculated exclusively from the training dataset and then applied to the test dataset. This prevents indirect information flow from the test data into the training process.

Additionally, data augmentation techniques were applied only to the training data, ensuring that the test data remained completely unseen and unbiased. No overlapping samples were used across validation or testing phases, maintaining strict separation between datasets.

Furthermore, careful monitoring of data pipelines was performed to ensure that there was no unintended mixing of data during batch generation or preprocessing stages. The use of separate data loaders for training and testing further reinforced this separation.

The model evaluation process was designed to simulate real-world scenarios, where unseen data is presented to the system. This approach ensures that the model's performance reflects its true ability to generalize rather than memorize patterns from the dataset.

These precautions ensure that the model evaluation reflects true generalization performance rather than artificially inflated results caused by data leakage. As a result, the reported accuracy and performance metrics provide a reliable and unbiased measure of the system's effectiveness.

4.6 Impact of Training Data Size on Classification Accuracy

The performance of the proposed hybrid CNN–Vision Transformer (ViT) model is significantly influenced by the size and diversity of the training dataset. A sufficiently large and varied dataset is essential for enabling the model to learn meaningful and generalizable features.

When the size of the training dataset is reduced, the model may fail to capture important variations such as differences in lighting conditions, camera quality, packaging textures, and subtle manipulation patterns present in real-world refund images. This limitation can negatively affect the model’s ability to distinguish between real and AI-generated images, thereby reducing overall classification accuracy.

A smaller dataset also weakens the model’s generalization capability, as it may not be exposed to diverse examples during training. As a result, the model may struggle when presented with unseen or complex inputs during testing or real-world deployment.

In contrast, larger and more diverse datasets enable:

- The CNN component to effectively learn fine-grained local features and artifacts
- The Vision Transformer component to capture global contextual inconsistencies across images

However, limited data increases the risk of overfitting, where the model memorizes training samples instead of learning general patterns. This leads to high training accuracy but poor performance on unseen data.

Moreover, insufficient data may result in class imbalance issues, where one class (real or fake) dominates the training process, leading to biased predictions. It can also reduce the model’s ability to detect rare or complex manipulation patterns, which are often critical in fraud detection scenarios.

To address these challenges, techniques such as data augmentation, transfer learning, and regularization can be employed. Data augmentation artificially increases dataset size by generating variations of existing images, while transfer learning allows the model to leverage knowledge from pretrained models. Regularization techniques help prevent overfitting and improve generalization. Overall, the results indicate that adequate and diverse training data is crucial for achieving high accuracy and reliable performance. Ensuring sufficient data representation is essential for deploying the system effectively in real-world refund fraud detection scenarios.

CHAPTER 5

CONCLUSION

5.1 Summary of Findings

The proposed system, FakeGuard AI, successfully demonstrates the practical implementation of a deep learning-based fake image detection framework integrated within a web-based application. The hybrid CNN–Vision Transformer (ViT) model effectively combines local feature extraction and global contextual understanding, enabling accurate classification of real and AI-generated images.

Experimental results indicate that the model achieves high classification performance, with an accuracy of approximately 93–94%, along with strong precision, recall, and F1-score values. The confusion matrix analysis further confirms minimal misclassification, highlighting the model’s reliability in distinguishing between authentic and fake images.

The system also exhibits stable training behavior, as observed from the loss and accuracy curves, which demonstrate smooth convergence and good generalization without significant overfitting. The integration of preprocessing, model inference, and result visualization ensures a seamless end-to-end workflow.

Additionally, the developed web interface provides a user-friendly platform for real-time image verification. Features such as instant prediction, confidence score visualization, and chatbot assistance enhance usability and accessibility, making the system suitable for both technical and non-technical users.

Overall, the project successfully bridges the gap between deep learning model development and real-world deployment, offering a scalable and practical solution for refund fraud detection and image authenticity verification.

5.2 Limitations and Future Scope

Despite its promising performance, the proposed system has certain limitations. At present, the model is limited to single-image classification, which restricts its ability to handle complex and dynamic fraud scenarios. The reliance on publicly available datasets may not fully represent the diversity of real-world refund claims, affecting robustness and generalization. Furthermore, the system depends solely on visual data, which may limit decision accuracy in some cases. From a deployment perspective, challenges related to scalability, computational efficiency, and real-time inference still need to be addressed. Additionally, the model may be vulnerable to adversarial attacks and intentionally manipulated inputs, which can impact reliability in practical applications .

To overcome these limitations, several improvements can be explored in future work. Extending the system to video-based analysis with temporal feature extraction could enhance its ability to detect fraud in dynamic content. Collecting larger, domain-specific, and more diverse datasets would improve model robustness and generalization. Incorporating multimodal inputs such as metadata, transaction history, and user behavior can further increase accuracy and reduce false predictions. Optimizing the system for scalability through cloud-based deployment and faster inference mechanisms is also essential for real-world applications. Strengthening the model against adversarial attacks should be prioritized to ensure reliability. Additionally, integrating a human-in-the-loop mechanism can help validate critical decisions in sensitive scenarios like financial transactions. Future enhancements may also include integration with e-commerce platforms for automated refund validation and continuous model retraining to adapt to evolving fraud patterns.

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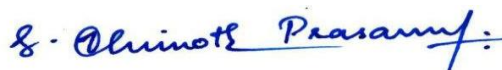
Name and Signature of all Batch members

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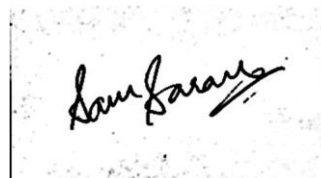
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